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The covariant origin of microforce balance in continua with coarse-grained scalar internal variables

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Highlights

- Scalar internal variables are recognized as metric-dependent observations.
- Spatial covariance of the First Law yields a unified master balance identity.
- Classical balances, microstress and microforce balance emerge from one structure.
- Observation covariance reveals the origin of the scalar microforce balance.
- The balance and thermodynamic principles admit a compact axiomatic organization.

Abstract

Microforce balance incorporates microstructural evolution into the balance structure in continuum theories with internal scalar variables, including plasticity, fracture, phase transitions, diffusion, and coupled multiphysics processes. It is usually introduced as an additional postulate, leaving unclear why it should belong to the balance structure of continuum mechanics. We show that the balance has a covariant origin in the spatial covariance of the First Law of Thermodynamics once the scalar internal variable is recognized not as a metric-independent primitive field, but as a metric-dependent coarse-grained observation of an underlying microstructural state. This covariant comparison between spatial observers yields a master identity in which the terms induced by the metric-dependent observation procedure identify the microstress and give the local microforce balance for the non-degenerate coarse-grained scalar variables considered here. Accordingly, within this class of continua, the microforce balance need not be introduced as an independent balance postulate. The remaining covariance components of the same identity recover mass balance, linear and angular momentum balances, and Doyle–Ericksen relations. For theories in

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First Law of Thermodynamics, and the Second Law of Thermodynamics. When the scalar internal variable represents conserved scalar content, one additional species balance is required. The present covariant derivation therefore provides a compact and geometrically unified organization of the balance laws and thermodynamic principles for first-gradient continua with coarse-grained scalar internal variables.

Keywords

Microforce balance; Spatial covariance; First Law of Thermodynamics; Coarse-grained scalar internal variables; Doyle–Ericksen relation

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Introduction

The motion of material points is the basic kinematic object of classical continuum mechanics, but many material responses cannot be described by placement, velocity, deformation, stress, and thermodynamic fields alone. Phase composition, damage, plastic rearrangement, defect content, concentration, and related microstructures may evolve independently of any additional spatial motion of the body while still affecting the material response. Internal-variable theories provide a continuum language for such processes by enriching the state with scalar or tensorial fields that represent coarse-grained information about the underlying microstructure. When the energy depends on these fields and on their spatial gradients, the same description also accounts for interfacial energy, nonlocal interactions, and internal length scales. This point of view underlies phase-transition theories (Landau, 1937, Ginzburg and Landau, 2009, Allen and Cahn, 1979), diffusion, phase-separation, and chemo-mechanical transport theories (Cahn and Hilliard, 1958, Cahn, 1961, Svendsen et al., 2018, Shanthraj et al., 2020, Bartels et al., 2021), variational and phase-field approaches to fracture (Francfort and Marigo, 1998, Ambrosio and Tortorelli, 1990, Bourdin et al., 2000, Miehe et al., 2010), and gradient plasticity and crystal plasticity (Aifantis, 1992, Fleck and Hutchinson, 1997, Fleck and Hutchinson, 2001, Gurtin, 2002, Gurtin and Anand, 2005). The present work is concerned specifically with continuum mechanics with scalar internal variables and their first gradients. Tensor-valued internal variables are not considered in the present development.

Classical continuum mechanics is organized by mass conservation, the balances of linear and angular momentum, and the First and Second Laws of Thermodynamics, which provide the axiomatic basis on which material-specific constitutive equations are built (Truesdell and Toupin, 1960, Truesdell and Noll, 2004, Coleman and Noll, 1974, Coleman and Gurtin, 1967). These principles separate the balance laws that every admissible motion must satisfy from the constitutive equations that specify stress, heat flux, and dissipative response for a particular material. Internal-variable theories enlarge this structure by adding fields that describe microstructural information not contained in the macroscopic motion of material points. When the internal state is spatially inhomogeneous, its exterior differential is the intrinsic one-

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rate, which introduces an action conjugate to that rate. A first-gradient dependence of the energy on the spatial differential of the scalar field also brings in a stress-like action conjugate to that differential (Germain, 1973, Mindlin, 1963). The evolution law is obtained only after the energetic action, the dissipative response, the restrictions imposed by the Second Law, and, in the conserved case, the transport balance for fluxes and sources have been specified. Although such a construction yields a closed evolution equation after the constitutive and kinetic choices have been made, it does not by itself separate the universal mechanical structure from the material-dependent part of the theory. In rational continuum mechanics, this separation is achieved by assigning stress a role in the balance laws before its constitutive form is specified, keeping the balance of actions conceptually distinct from the material constitutive behavior. The corresponding issue for internal-variable continua is whether the actions conjugate to the rate of the internal variable and to the rate of its spatial differential can be organized in the same way, through a local balance fixed before constitutive laws and dissipative kinetics are chosen.

Gurtin's microforce theory answers this question by treating the actions associated with internal-variable evolution as quantities that satisfy a local balance, in analogy with stress in the macroscopic theory (Gurtin, 1996). The theory postulates a local balance among the microscopic actions, thereby assigning definite mechanical meanings to the microstress, the internal microforce, and the external microforce supply. The important point is the separation between the balance of microscopic actions and the constitutive laws of a particular material. The microforce balance fixes the local relation among the rate-conjugate actions, whereas constitutive equations and dissipative kinetic assumptions determine the material response. This separation places Cahn–Hilliard and Ginzburg–Landau/Allen–Cahn-type kinetics within a common thermomechanical theory (Gurtin et al., 1996, Gurtin, 1996, Gurtin, 2003). The same structure underlies microforce-based theories of gradient plasticity and finite-deformation crystal plasticity (Gurtin, 2002, Gurtin and Anand, 2005, Gurtin, 2008, Wulfinghoff, 2017), as well as phase-field and gradient-damage formulations of fracture (Duda et al., 2015, Duda et al., 2018, Geelen et al., 2019). Related formulations include nonlocal granular rheology (Henann and Kamrin, 2014), variational couplings of Cahn–Hilliard-type diffusion with crystal plasticity and Allen–Cahn-type phase transformations in dissipative solids (Nagaraja et al., 2023, Nagaraja and Antretter, 2024), and multiphysics phase-field models coupling diffusion, phase transformations, deformation, and fracture (Zhao et al., 2026).

An alternative and closely related framework is based on the principle of virtual power. In classical continuum mechanics, the virtual-power identity, together with the requirement that internal virtual power vanish on rigid virtual motions, constitutes a weak integral form of the balances of linear and angular momentum and the associated traction conditions. Germain recast this principle as a constructive foundation for higher-gradient and microstructured continua (Germain, 1973, Del Piero, 2009, Epstein and Smelser, 2020). The construction begins by specifying the admissible virtual kinematics and the power expenditures of the internal and external actions, after which equality of these powers is imposed for every material part and every admissible virtual field. Frémond and Nedjar (Frémond and Nedjar, 1993, Frémond and Nedjar, 1996) applied this construction to damage by including

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transitions, shape-memory materials, and non-smooth thermomechanical processes (Frémond and Shitikova, 2002). In gradient plasticity, Gurtin likewise used the principle of virtual power to introduce microforce systems work-conjugate to slip rates, plastic-strain rates, and their gradients, leading to the corresponding microforce balances and microscopic boundary conditions (Gurtin, 2002, Gurtin, 2003, Gurtin and Anand, 2005).

Once the microscopic kinematics and their power-conjugate actions have been fixed, integration by parts separates the volume and boundary contributions to the virtual-power identity, and the arbitrariness of the virtual fields yields the local microforce balances in the interior and the microtraction conditions on the boundary. Conversely, the local relations recover the virtual-power identity, so the two representations are mathematically equivalent under the usual regularity assumptions. A local balance theory takes the balance of microscopic actions as an initial mechanical postulate. A virtual-power theory begins instead with the admissible microscopic virtual kinematics, the corresponding internal and external power functionals, and their equality, from which the local balance structure follows by localization.

The preceding discussion indicates that the local microforce balance and the virtual-power principle may be regarded as two representations of the same underlying physics. The present work examines the balance structure through its local formulation. The local microforce balance is introduced by analogy with the macroscopic force balance (Gurtin, 1996, Gurtin, 2000, Gurtin et al., 2010). This analogy places the evolution of internal variables within a balance structure comparable to that governing macroscopic motion, but it also reveals the central difficulty addressed in this paper. The classical force balances are associated with the spatial motion of material bodies. A scalar internal variable, by contrast, is ordinarily introduced as an additional state descriptor rather than as a motion in physical space. It is therefore not evident why such a descriptor should carry a balance law of its own.

The scalar microforce balance raises an axiomatic question different from the one encountered in the macroscopic theory, but the latter provides a useful geometric insight. Balance laws whose integral forms appear natural and self-contained in Euclidean space are not intrinsically defined once the spatial geometry is made explicit. In particular, the usual Euclidean integral form of the balance of linear momentum relies on the affine identification of vectors attached to different spatial points, while the balance of angular momentum relies on position vectors, the cross product, and the construction of moments about an origin. A general spatial manifold carries no such Euclidean structure intrinsically. The First Law of Thermodynamics has a different geometric status, because it equates two scalar quantities, the rate of energy of a part and the power supplied to that part. It can therefore serve as an intrinsic energy statement whose spatial covariance generates the local momentum balances, rather than requiring the integral balances of linear and angular momentum to be introduced as independent axioms.

The idea that the macroscopic balance laws can be recovered from the invariance or covariance of an energy balance is already present in the work of Green, Rivlin, and

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1964). Marsden and Hughes replaced rigid observer changes by spatial diffeomorphisms of the ambient space, recovering the classical balances together with the Doyle–Ericksen stress relation from covariance of the energy balance (Marsden and Hughes, 1994). The Doyle–Ericksen relation, originally introduced as a stress–metric connection in nonlinear elasticity (Doyle and Ericksen, 1956), was subsequently clarified in geometric and material forms through the rotated-stress and spatial/material covariance theories (Simo and Marsden, 1986, Yavari et al., 2006). These results give covariance a constructive role, not merely as a requirement on the form of the equations, but as the mechanism through which the energy balance generates local balance laws. This motivates the corresponding question for internal-variable continua. The question is whether the spatial covariance of the First Law of Thermodynamics can explain why the scalar microforce balance belongs to the continuum balance structure, rather than requiring it to be introduced as an additional postulate by analogy with the macroscopic balances.

The corresponding question for internal variables has been approached from both geometric and thermodynamic directions. One geometric line of work enlarges the kinematic description of the body by assigning each material point an additional substructural descriptor, often valued in an independent manifold of microstructural states. This approach works well when the microstructure carries its own kinematic representation, but it does not by itself resolve the scalar internal-variable problem considered here. In such theories, covariance of an enlarged energy or power statement can yield both macroscopic and substructural balances (Yavari and Marsden, 2009). It is natural for director-like, orientational, or manifold-valued microstructures, because the substructure itself supplies a kinematic object on which changes of observer or representation can act nontrivially. The scalar variables used in these theories are usually not placements into an independent microstructure manifold. If they are treated as primitive scalar fields, a spatial change of observer only relabels their spatial representation. No geometric mechanism is then available from which a Gurtin-type scalar balance can be extracted.

The origin of the balance has also been sought through thermodynamic admissibility. Generalized Coleman–Noll procedures require the Clausius–Duhem inequality to hold over an enlarged class of admissible processes and can recover balance laws for continua with an underlying energy function, such as hyperelastic materials (Sadik and Yavari, 2025). The difficulty arises for scalar internal variables whose evolution is genuinely dissipative. In Cahn–Hilliard, Ginzburg–Landau, and related phase-field models, the internal variable is not merely an additional state parameter whose rate can be freely varied in a Coleman–Noll separation. Its rate belongs to the dissipative structure of the theory, rather than to an arbitrary class of process variations. If this rate is treated as arbitrary, the dissipation inequality forces the associated thermodynamic driving force to vanish, suppressing the dissipative evolution that the theory is meant to describe. If a dissipative mechanism is prescribed instead, the Coleman–Noll separation no longer extracts the microforce balance from the dissipation inequality alone. The admissible-process route therefore cannot by itself identify the origin of the scalar microforce balance in dissipative internal-variable theories.

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This shows that the scalar microforce balance need not be postulated in Gurtin's original form, but can be recovered by imposing a covariance requirement on the Second Law. The derivation, however, rests on a strong additional admissibility axiom imposed on the transformed inequality. When the inequality written by one observer is pulled back to the representation of another observer, the additional pullback contribution is required to be non-positive and to vanish for rigid observer changes. Thus the microforce balance is no longer inserted as an independent balance axiom, but is recovered through a nontrivial covariance requirement imposed on the transformed Second-Law inequality. The limitation is therefore not the result itself, but the origin assigned to it, since the balance remains tied to an admissibility condition on that transformed inequality.

The covariance tradition of Green–Rivlin–Naghdi and Marsden–Hughes suggests a different axiomatic organization. Balance laws are most naturally tied to exact energy and power statements, where power-conjugate actions are identified before constitutive or kinetic restrictions are imposed. The First Law of Thermodynamics has precisely this character, since it equates the rate of total energy of every part with the power supplied to that part. The Second Law of Thermodynamics plays a different role by restricting admissible constitutive behavior and dissipative mechanisms after the relevant power expenditures and balance relations have been specified. This separation becomes essential for internal-variable theories in which the rate of the scalar field carries dissipative microstructural kinetics. In such theories, the dissipation inequality should constrain the dissipative kinetics associated with the internal-variable rate, rather than serve as the source from which the microforce balance is forced.

This motivates seeking, for the scalar first-gradient class considered here, a covariant origin of the microforce balance in the spatial covariance of the First Law of Thermodynamics. This keeps the balance relations tied to power conjugacy, while the dissipation inequality retains its classical role in constitutive behavior and dissipative kinetics. It also raises the question whether the scalar microforce balance belongs to the same covariance–balance correspondence that yields mass balance, linear and angular momentum balances, and the Doyle–Ericksen relation from observer covariance of the First Law of Thermodynamics.

The central point is that a coarse-grained scalar internal variable appears as a scalar-valued continuum field only after an observation procedure has converted microscopic information into a continuum representation. This observation procedure is metric-dependent, because the finite neighborhood over which microscopic information is resolved has its size, shape, volume weighting, and notion of proximity determined by the spatial metric. The resulting scalar representation therefore inherits metric dependence, even though the continuum field obtained after coarse graining is scalar-valued. If this observational origin is suppressed and the variable is treated as an ordinary scalar field, a spatial change of observer merely relabels its spatial representation and generates no independent geometric variation from which a microforce balance could arise.

A spatial change of observer consequently affects not only the representation of motion, stress, volume, and surface measure, but also the metric argument of the

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coarse-grained internal representation, together with the usual observer changes of the macroscopic fields and geometric measures. The scalar field therefore enters the covariance of the First Law of Thermodynamics through the metric dependence of the observation procedure that defines it.

With this metric-dependent observation included, the spatial covariance of the First Law of Thermodynamics yields a master identity. This identity contains a covariance component associated with the metric-dependent observation procedure that defines the coarse-grained scalar state. For the non-degenerate coarse-grained scalar variables considered here, this component separates into a first-gradient relation that identifies the microstress and a scalar relation that gives the local microforce balance. The same identity also carries the classical covariance components. The quadratic dependence on the observer velocity yields mass balance, translational covariance yields the balance of linear momentum, rotational covariance yields the balance of angular momentum, and the metric-changing part yields Doyle–Ericksen relations.

The microforce balance is therefore not appended as an independent balance postulate to the macroscopic theory. In the present class of continua, the metric-dependent coarse-grained observation enters the comparison of the First Law between spatial observers, and spatial covariance yields a master identity containing both the classical and observation-induced contributions. After localization, the local non-degeneracy condition permits separation of the observation-induced responses, from which the microstress identification and the local microforce balance follow. The Second Law of Thermodynamics remains an independent principle that restricts constitutive response and dissipative kinetics rather than supplying these balance relations. Accordingly, for a non-conserved scalar internal variable, the organization of the balance and thermodynamic principles considered here comprises the First Law of Thermodynamics, spatial covariance of the First Law, and the Second Law of Thermodynamics. When the scalar variable represents conserved content, a species balance must additionally be supplied. The present covariant derivation thus provides a compact and geometrically unified organization of the balance laws and thermodynamic principles for first-gradient continua with coarse-grained scalar internal variables.

The paper is organized as follows. Section 2 establishes the geometric setting, introduces the spatial and material first-gradient representations of the coarse-grained scalar state, and states the First Law of Thermodynamics. Section 3 formulates spatial changes of observer and the metric-dependent coarse-graining procedure. Section 4 derives the master covariance identities, obtains the classical balances, and uses the local non-degeneracy condition to separate the Doyle–Ericksen relation, the microstress identification, and the scalar microforce balance. The covariance construction is then extended to anelastic continua. Section 5 specializes the theory to Ginzburg–Landau and Cahn–Hilliard models and to scalar-gradient finite plasticity. Section 6 discusses the physical meaning, scope, and relation to existing formulations, and Section 7 summarizes the main conclusions.

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Geometry and the first law of thermodynamics

The theory is formulated on smooth oriented Riemannian manifolds so that the First Law of Thermodynamics does not rely on auxiliary Euclidean structure. A motion maps the material body into physical space, whose metric determines the volume and area measures, unit normals, tensorial pairings, fluxes, and power expenditures associated with each current part. We first establish this geometric setting and the kinematic conventions used throughout the paper. The scalar internal variable is then ...

Spatial changes of observer and metric-dependent coarse graining

So far, the First Law of Thermodynamics has been written in the current spatial representation, where volume, area, normal direction, rates, and power expenditures are measured by the spatial metric g . In this representation, the coarse-grained internal variable appears as a spatial scalar field c . Its constitutive first-gradient state may be chosen as either the spatial first jet (c, p) , with $p = dc$, or the material first jet (C, P) , with $C = c \circ \varphi_t$ and $P = dC$. Spatial covariance requires comparing the ...

Spatial covariance of the first law of thermodynamics

The principle of covariance has its classical expression in general relativity, where fundamental laws are formulated independently of the coordinates used to represent spacetime (Einstein, 1916). In continuum mechanics, the corresponding requirement is spatial. Under a change of observer, the motion, the spatial metric, and the fields entering the energy and power are replaced by their transformed representatives, while the energy balance retains its invariant form and continues to describe ...

Representative internal-variable models

We now specialize the general covariance structure to representative continuum models with internal variables. Ginzburg–Landau and Cahn–Hilliard models provide important examples of non-conserved and conserved scalar internal variables, respectively, while finite plasticity provides a complementary setting in which the scalar internal variable is coupled to an anelastic deformation through F^a . The purpose of these examples is to show that, under standard constitutive assumptions, the microforce ...

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variables are generated by metric-dependent coarse-graining procedures with locally non-degenerate observation responses. In standard continuum mechanics with internal variables, this balance is usually introduced as a postulate for microscopic actions (Gurtin, 1996). Such a postulate assigns precise mechanical roles to ...

Conclusion

For the Cauchy-type first-gradient continua with non-degenerate, metric-dependent coarse-grained scalar internal variables considered here, spatial covariance of the First Law of Thermodynamics provides a covariant origin of the local microforce balance. The scalar field is not a primitive continuum variable, but a continuum reading of underlying microscopic or mesoscopic information through a metric-dependent spatial observation process. A spatial change of observer therefore acts not only on ...

CRediT authorship contribution statement

Chenlong Zhao: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Guisen Liu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Yao Shen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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